
Translate Needs To Integration Requirements

Given a use case, identify performance needs (volumes, response times, latency) and propose appropriate integration solutions that will meet business requirements.

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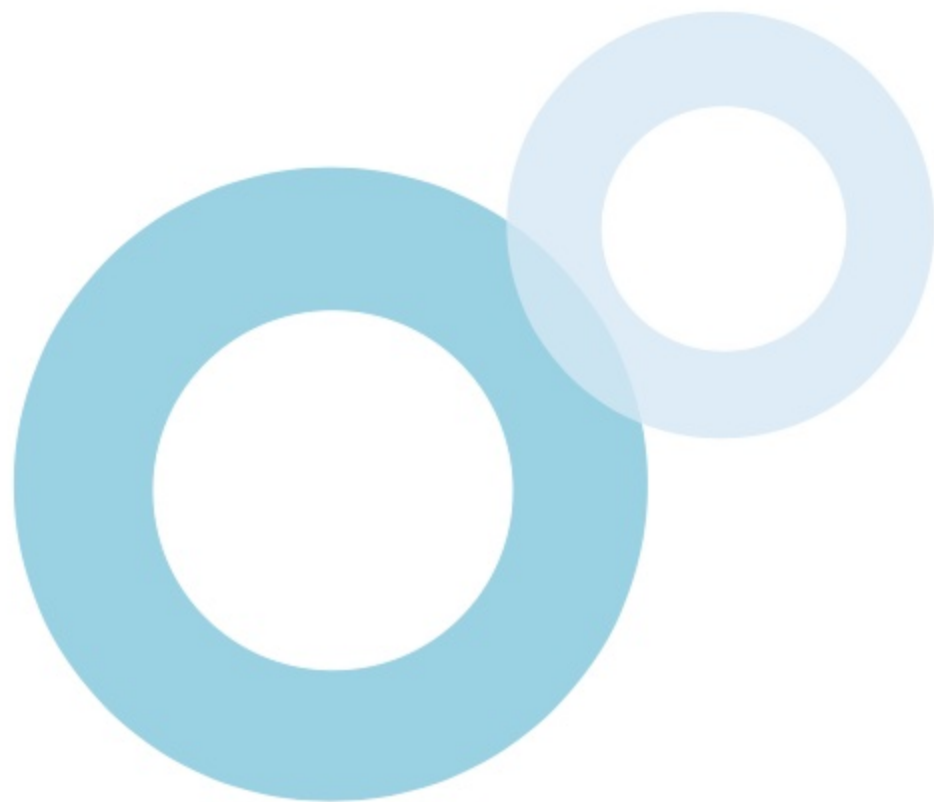
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After studying this topic, you should be able to:

-  Identify the considerations related to data volumes, response times, and latency for various integration patterns.
-  Identify appropriate integration solutions for the performance needs in the given business requirements.



Introduction

The choice of the integration pattern for an integration between Salesforce and a remote system may depend on the required **data volumes, response times, and latency**. For example, an integration based on the **'Remote Process Invocation - Request & Reply'** pattern is suitable for **small volume, real-time activities**.

REST/SOAP API can be invoked from a remote system to create, update or delete **200 records** at a time. If a data load for data synchronization with a remote system requires processing of **more than 2,000 records**, the use of **Bulk API 2.0** can be considered. **Synchronous Apex callouts** are suitable for **small volume, real-time** activities. **Salesforce Connect** can be used to give users real-time access to data.



Overview

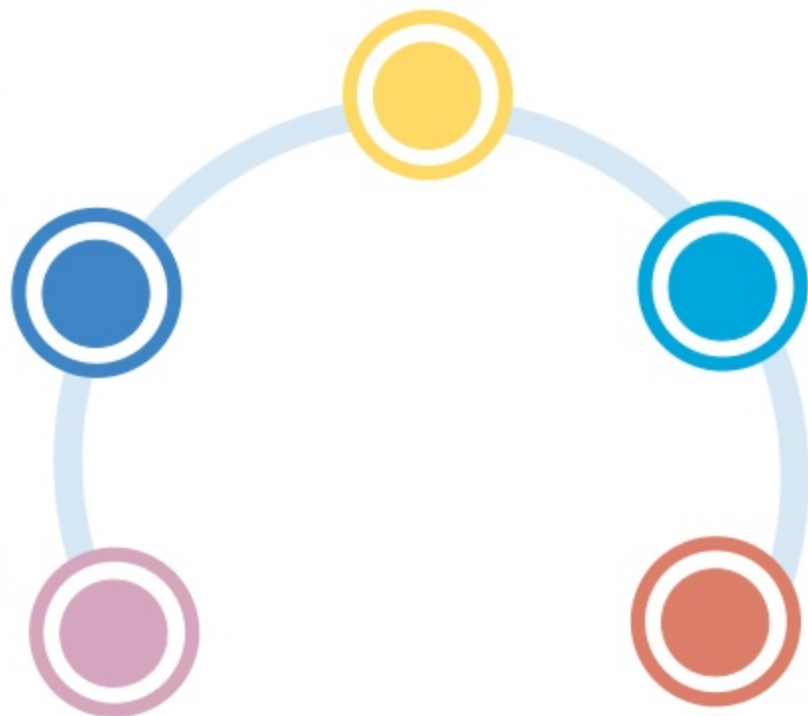
Bulk API 2.0 can be used for processing **more than 2,000 records** at a time. **Data segmentation** can be used to avoid data contention during business hours.

The data volume considerations for an integration based on the **'Remote Process Invocation - Fire & Forget'** pattern depend on the chosen solution.

An integration based on the **'Remote Process Invocation - Request & Reply'** pattern is suitable for **small volume, real-time activities**. If the remote server has **high latency**, **asynchronous Apex callouts** should be used.

Size limits are applicable to inserting or updating **Blob data** in standard objects. The **maximum file size** for uploads is **2 GB** for the **ContentVersion** object and **500 MB** for all other standard objects.

When **data virtualization** is required, **synchronous Apex callouts** are suitable for **small volume, real-time activities**. **Salesforce Connect** can be used for real-time access.



Integration Performance Needs

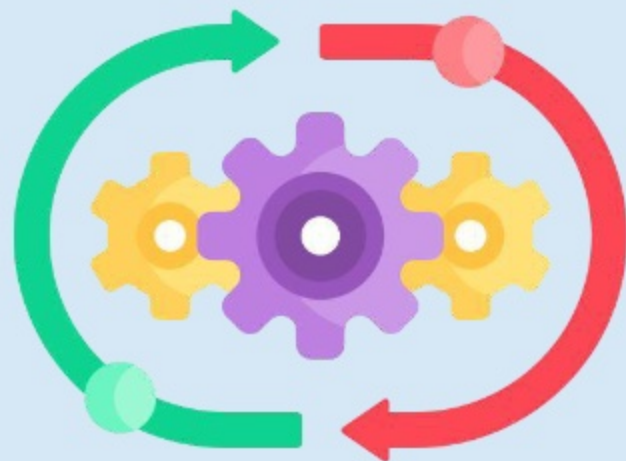
The choice of the **integration pattern** for an integration between Salesforce and a remote system may depend on the required **data volumes**, **response times**, and **latency**.

❖ REMOTE PROCESS INVOCATION - REQUEST & REPLY

An integration based on the 'Remote Process Invocation - Request & Reply' pattern is suitable for **small volume, real-time activities** since the timeout is short and the request or response has a maximum size.

It can be used for **synchronous** calls to the remote system when the end user requires an immediate response. The remote process should be executed in a timely manner to conclude within the **configurable timeout of up to 120 seconds**.

Also, the remote system should have a **low latency response**. If it poses a risk of high latency response, then an **asynchronous callout, also known as continuation**, should be used to avoid hitting the synchronous Apex transaction governor limits. Although a continuation is **executed asynchronously**, its implementation still falls under the request & reply pattern since a **reply is expected** through a callback.



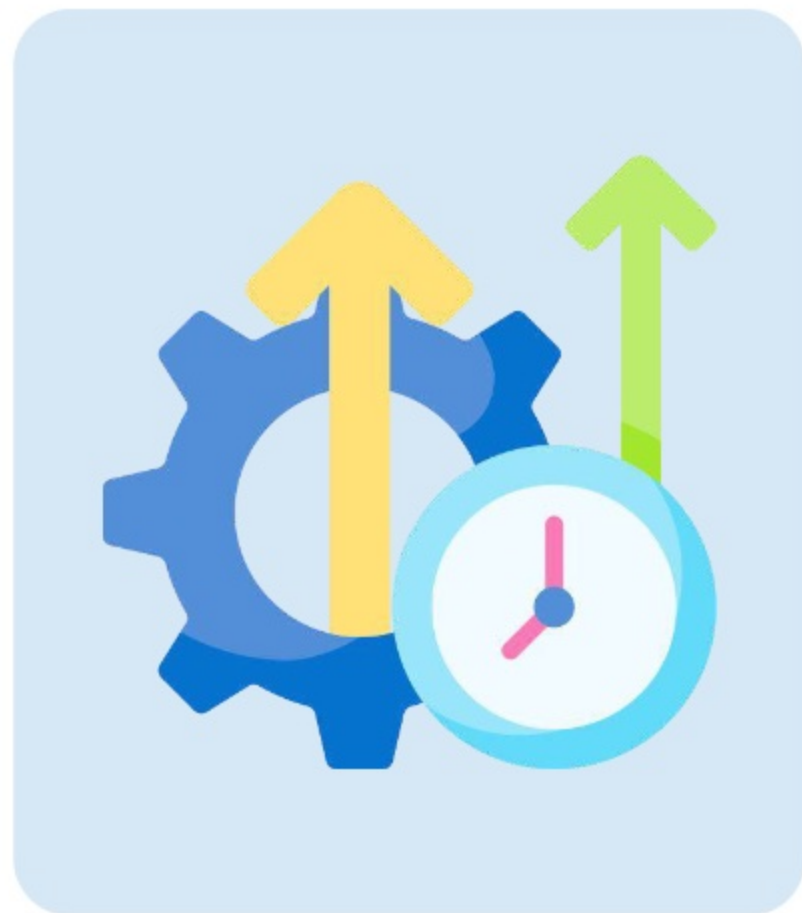
Integration Performance Needs

The choice of the **integration pattern** for an integration between Salesforce and a remote system may depend on the required **data volumes**, **response times**, and **latency**.

❖ REMOTE PROCESS INVOCATION - FIRE & FORGET

Various considerations apply to an integration based on the 'Remote Process Invocation - Fire & Forget' pattern. The **data volume considerations** depend on the chosen solution. For example, the maximum platform event message size is 1 MB, which can be hit if the event object has hundreds of custom fields or many long text area fields.

This pattern is suitable when Salesforce **does not need to wait** for a call's successful completion. **Timeliness** is not an important factor since the control is handed back to the client immediately or after **positive acknowledgement** of a successful hand-off to the remote system.



Integration Performance Needs

The choice of the **integration pattern** for an integration between Salesforce and a remote system may depend on the required **data volumes**, **response times**, and **latency**.

❖ BATCH DATA SYNCHRONIZATION

In batch data synchronization, if a data load requires processing of **more than 2,000 records**, the use of **Bulk API 2.0** can be considered. **Change Data Capture** can be used to receive **near real-time changes** of Salesforce records and synchronize them in an external data store.

Interfaces must be designed to ensure that all the batch processes complete in a **designated batch window**. **Batch processing during business hours** can result in **data contention**. **Data segmentation techniques**, such as the use of **record types** and other **filter criteria**, can be used to avoid data contention.



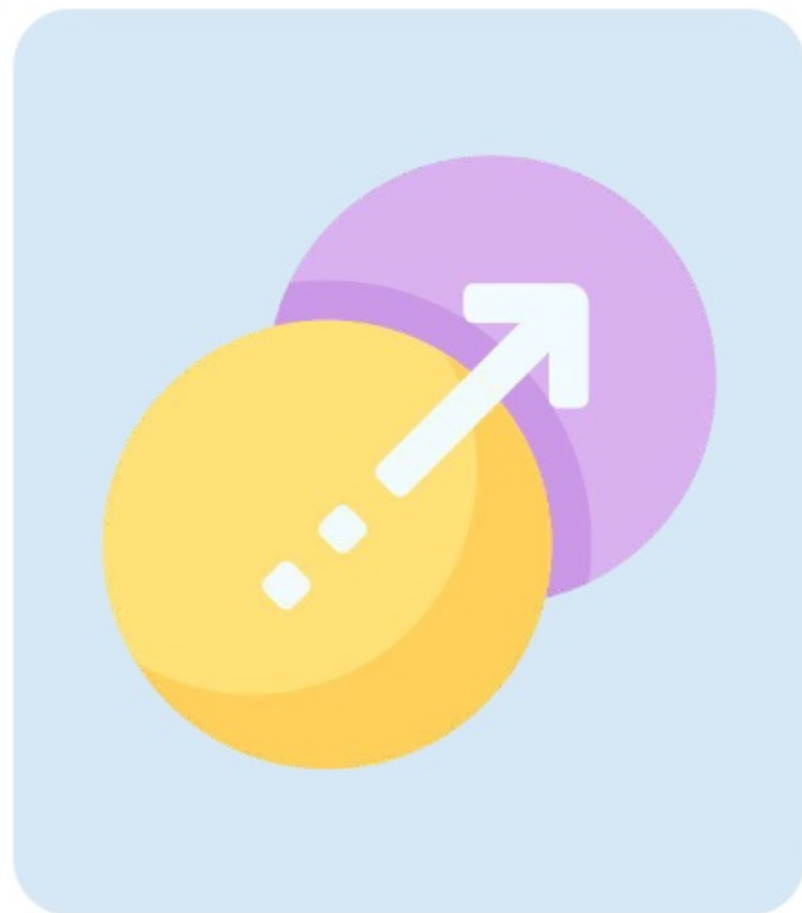
Integration Performance Needs

The choice of the **integration pattern** for an integration between Salesforce and a remote system may depend on the required **data volumes**, **response times**, and **latency**.

❖ REMOTE CALL-IN

REST or SOAP API can be invoked from a remote system to create, update or delete **200 records** at a time. **Bulk API 2.0** should be used for processing **more than 2,000 records** at a time. Also, up to **2,000 rows** can be returned in a **query()** or **queryMore()** call. **Size limits** are also applicable to **inserting or updating Blob data** in standard objects. The **maximum file size** for uploads is **2 GB for the ContentVersion object** and **500 MB for all other standard objects**.

REST and **SOAP APIs** are **synchronous**, which means that the remote system waits until it receives a response from Salesforce. **Session and query timeouts** are applicable to them. The session times out if there's no activity. Each SOQL query has a timeout limit of **120 seconds**.



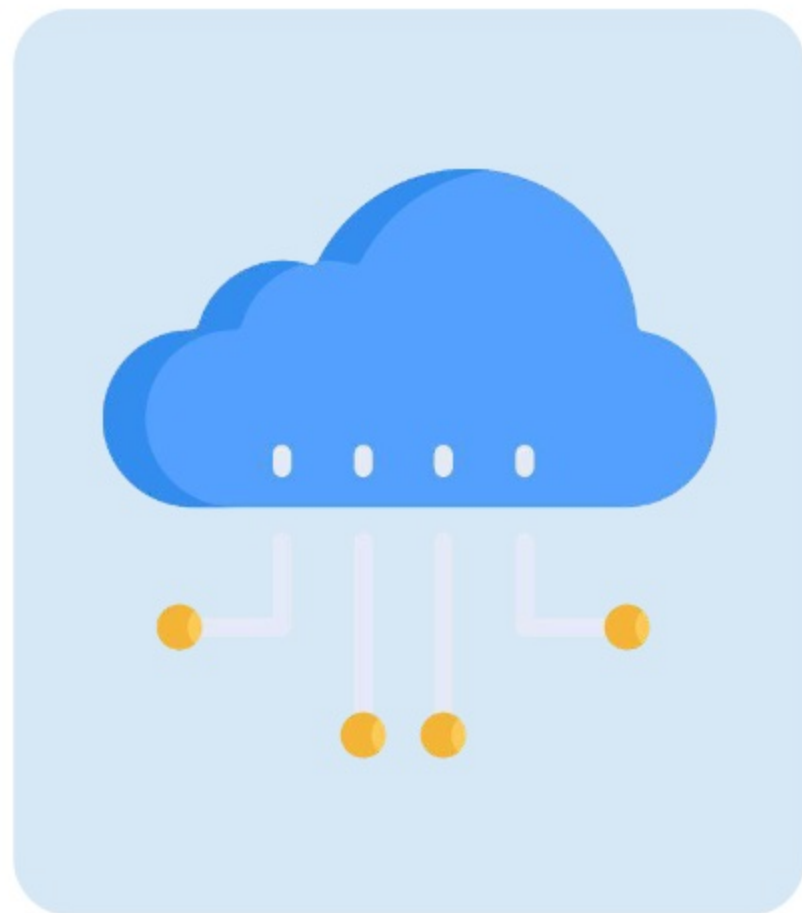
Integration Performance Needs

The choice of the **integration pattern** for an integration between Salesforce and a remote system may depend on the required **data volumes**, **response times**, and **latency**.

❖ DATA VIRTUALIZATION

When data virtualization is required, **synchronous Apex callouts** are suitable for **small volume, real-time** activities.

For **Apex callouts**, the request must be executed in a timely manner within the configurable timeout of up to **120 seconds**. When **real-time access** is required, **Salesforce Connect** can be used.



Performance

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Performance

System performance refers to the system's processing capacity (**throughput**) and its responsiveness to requests and demands (**latency**). Performance can be improved through **throughput optimization**.

❖ THROUGHPUT OPTIMIZATION

Throughput refers to the **number of concurrent requests**, which translates to **workload**, that an application can complete in a given time.

❖ WORKLOAD FACTORS

There are three factors that can be considered that contribute to workload in the system: **number of transactions**, **number of concurrent users**, and **complexity of the requests**.



Throughput Optimization

Salesforce recommends throughput optimization using any of the **following techniques**.



ASYNC PROCESS

Processing requests asynchronously can help **minimize potential bottlenecks**, but may also contribute to a **decrease in response times**.



BULK PROCESS

Bulkification of DML operations, callouts, and other tasks **reduces overhead**.



PROPER QUERIES

SOSL should be used for **text searches** in records while SOQL is used for **efficient retrieving** of records.



PLATFORM CACHE

The Platform Cache can be used to **cache sessions** and **org data** for reduced computing requirement.

Latency Optimization

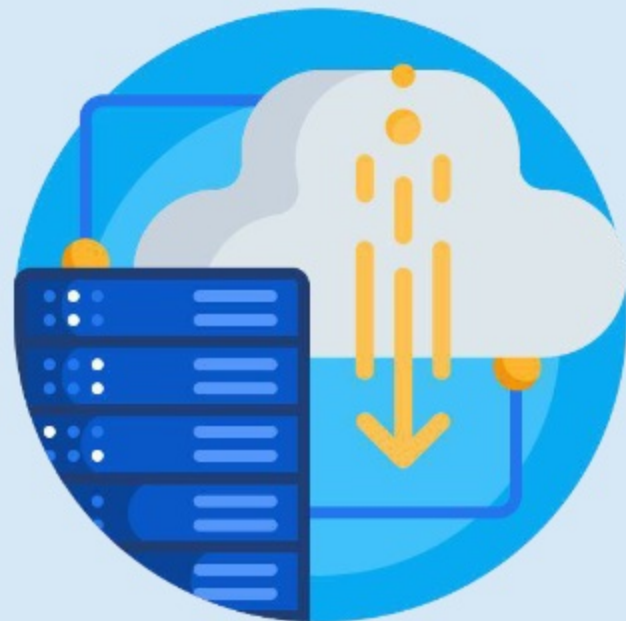
Performance of a system can also be improved by reducing latency, which is a **time measurement** of how fast a system is able to **complete an execution path**.

❖ THROUGHPUT OPTIMIZATION

Improving system throughput will also help reduce the latency of the system. When a system always has the resources to handle additional workload, requests can be processed much faster.

❖ PERCEIVED PERFORMANCE

Another dimension of latency that should be considered is what is called as “perceived performance” or how seemingly responsive the system is based on the end user’s perspective.



Reducing Latency

Salesforce recommends using any of the **following techniques** for optimizing latency.



OPTIMIZE REPORTS

A report should only contain the **necessary data** intended for a **specific audience** to serve a **single purpose**.



OPTIMIZE FILTERS

Filter logic should use **equal/not equal** instead of **contains/does not contain** and filtering by **formula fields** should be avoided.



SIMPLIFY SHARING

Reducing the complexity of **sharing and visibility** configurations on Salesforce org data helps reduce latency.



OPTIMIZE INTERFACE

Lightning web components, Lightning Data Service, and client-side **data sorting/filtering** helps reduce latency in custom UIs.

Timing, Volume, and Scale

The following **factors** can determine the capability of an integration tool in enhancing **system performance**.



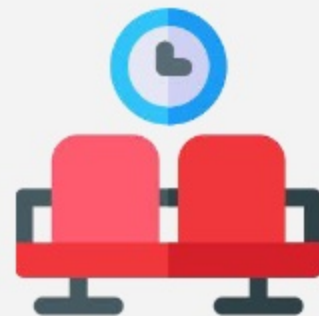
TIMING

Ability to communicate **synchronously** or **asynchronously**



VOLUME

Ability to efficiently work with **large volumes of data** or process data in bulk



OBJECT

Ability to process **multiple objects** in a single transaction or a **single object** only

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Smart Solutions is a startup company that sells smart home devices such as smart cameras, lights, door locks, doorbells, etc. It sells products to both individuals and businesses and uses Salesforce for managing records such as accounts, contacts, and opportunities.

In addition, it uses other enterprise systems to manage other types of data. For instance, an enterprise resource planning (ERP) system is used for storing orders and invoices. A human resource information system (HRIS) stores data about employees. These systems need to be integrated with Salesforce based on various performance needs. The solution architect needs to recommend appropriate integration solutions.

Requirements & Solutions

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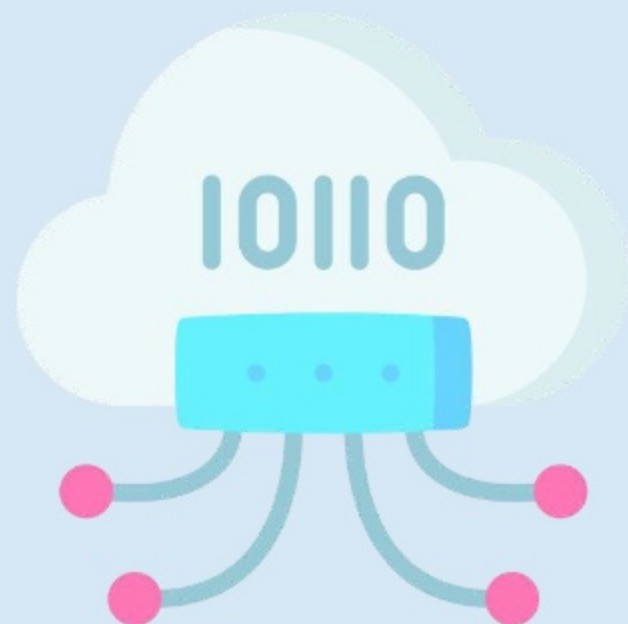


Requirement 1



SCENARIO

The sales reps of the company use Salesforce for opportunity management. They use an ERP system for accessing orders and related invoices. Once an opportunity is won, an order and a related invoice are created in the ERP system via an existing integration automatically. A custom user interface is required in Salesforce to allow sales reps to access and update orders in the ERP system. A typical sales rep would need to view or edit one order at a time in real-time and require immediate response from the ERP system. A web service has been defined in the ERP system for this use case. The solution architect needs to recommend a suitable solution for this requirement.



Solution 1



SOLUTION

The given requirement can be met using a synchronous integration solution since a sales rep would need to access or update the order data in real-time and require immediate response from the ERP system. A **'Remote Process Invocation - Request & Reply'** integration, such as a **synchronous Apex REST or SOAP callout** invoked from a **Lightning web component** can be used. It is suitable for **small volume, real-time activities** when the call to the remote system should be a synchronous request-reply.

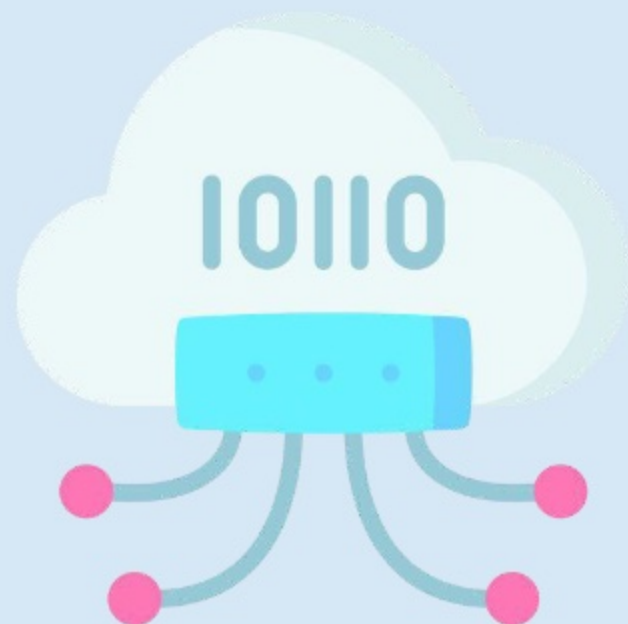


Requirement 2



SCENARIO

The support reps use an external system for case management. The support director wants to ensure that customer information in the external system is always up-to-date. When a Salesforce user updates an account or contact in Salesforce, the corresponding account or contact in the external system should be updated automatically. The user who performs the update does not need to wait for a response from the external system. Only a few custom fields have been defined on the Account and Contact objects in Salesforce.



Solution 2



SOLUTION

The requirement can be met by using an integration solution based on the **'Remote Process Invocation - Fire & Forget'** pattern, such as a **record-triggered flow that publishes a platform event message** or **sends an outbound message**. Since **Account** and **Contact** objects have only a few custom fields, the maximum platform event message size of **1 MB** should be sufficient for this use case.

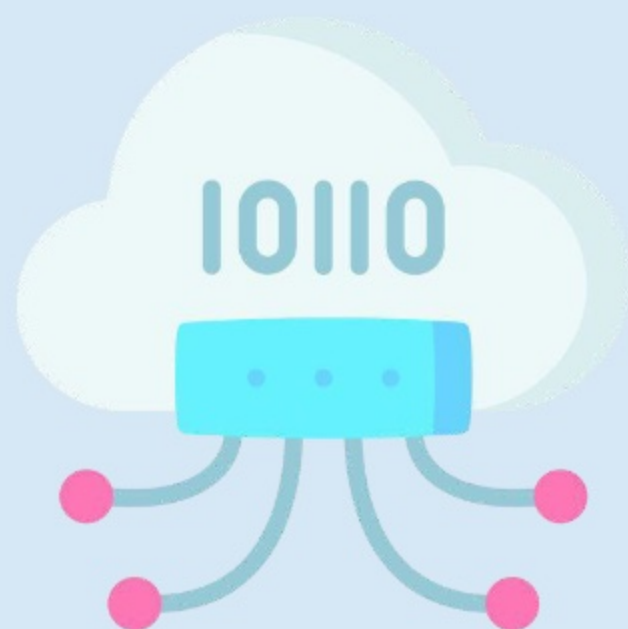


Requirement 3



SCENARIO

A developer needs to create a Lightning web component that allows the HR department to update data in the company's external HRIS (Human Resource Information System). Up to 100 HR users are expected to use the component at the same time, that will typically result in multiple concurrent long-running requests, which are transactions that run for more than 5 seconds, due to the complex processes required in the update operation. The architect must recommend a suitable solution for this requirement.



Solution 3



SOLUTION

Callouts are not included in the the long-running request limit. However, to ensure that the simultaneous actions performed by the HR users do not exceed the “**default**” **limit of 10** for **concurrent long-running synchronous requests**, asynchronous requests can be used to exempt the operation from this limit.

Salesforce uses **scaled license-based limits** for concurrent long-running synchronous requests. For example, if an org has **1,000 licenses** or **less**, the limit is **10** (min). If it has **2,000 licenses**, the limit is **20**. If it has **3,000** licenses, the limit is **30**, and so on. However, if it has **5,000** licenses or **more**, the limit is **50** (max).

Salesforce recommends using **continuations** when working with callouts for improved performance over regular callouts in asynchronous operations. The response of an asynchronous callout in a continuation is returned through a **callback method**.

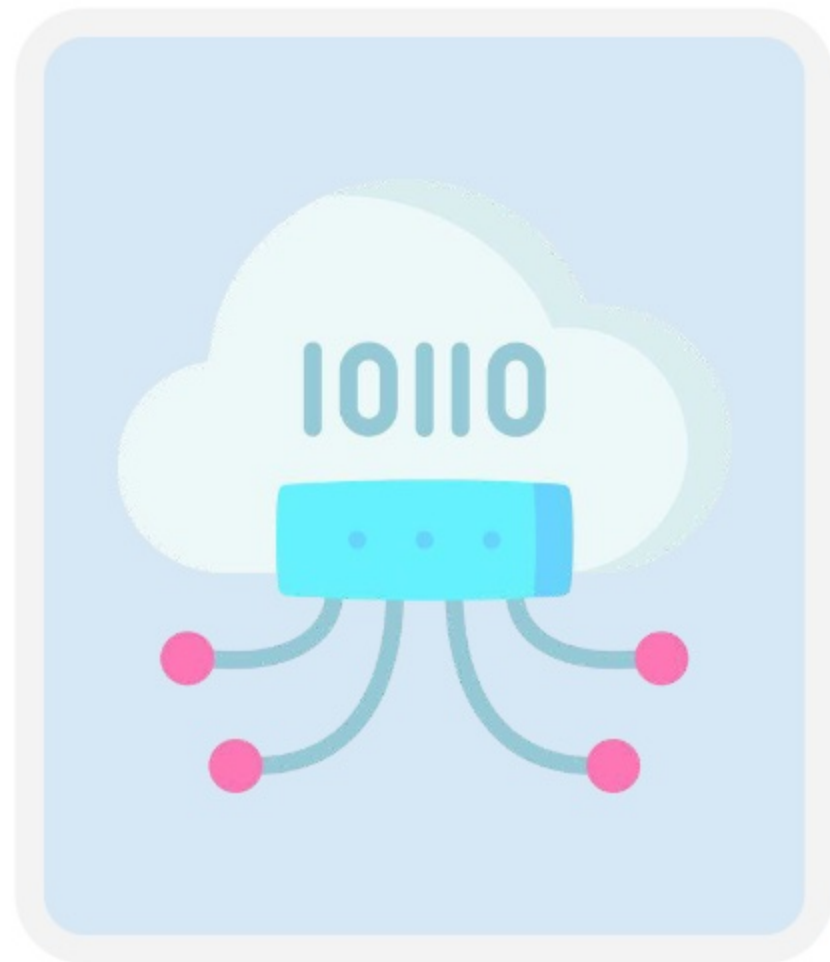


Requirement 4



SCENARIO

Thousands of orders placed by customers are stored in the ERP system. Certain sales users who can only access Salesforce need the ability to view and edit orders that exist in the ERP system. The ERP system supports the OData protocol and a sales rep would typically need real-time access to one order at a time. A suitable solution is required for this use case.



Solution 4



SOLUTION

Salesforce Connect can be used to give Salesforce users real-time access to data from external sources, such as the ERP system in the given requirement. Since the ERP system supports the **OData protocol**, using Salesforce Connect is better than using synchronous REST/SOAP callouts. The **OData 2.0/4.0/4.01 adapter** can be selected while setting up an **external data source** for this use case.



Learn More



[Integration Patterns and Practices](#)